Document Identifier: **PMC8600397**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: Mouse xenograft models and treatments All animal care procedures and experiments were conducted in accordance with the Animal Research: Reporting of In Vivo Experiments (ARRIVE) guidelines (22) and approved by Ethics Committee of Animal Experiments of the Hebei General Hospital (approval no. 20200147), Shijiazhuang, China).
Euthanasia Agents: At the end of the experiment, the mice were euthanized via CO2 inhalation (50 of the chamber volumemin), and the tumor tissues were collected and weighed.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
BALBc NUNU nude mice (male 8 weeks old body weight 20-30 g 40 mice in total) were purchased from Hebei Medical University and housed in the Experimental Animal Facility (five mice per cage) of the hospital under standard laboratory conditions (18-23degC 40-60 humidity 12-h lightdark cycle) with unlimited access to food and water.
Subject Demographics
Age: BALBc NUNU nude mice (male 8 weeks old body weight 20-30 g 40 mice in total) were purchased from Hebei Medical University and housed in the Experimental Animal Facility (five mice per cage) of the hospital under standard laboratory conditions (18-23degC 40-60 humidity 12-h lightdark cycle) with unlimited access to food and water.
Weight: BALBc NUNU nude mice (male 8 weeks old body weight 20-30 g 40 mice in total) were purchased from Hebei Medical University and housed in the Experimental Animal Facility (five mice per cage) of the hospital under standard laboratory conditions (18-23degC 40-60 humidity 12-h lightdark cycle) with unlimited access to food and water.
Randomization
The mice (20group) were randomly assigned into the following treatment groups: i) Mice injected with regular LNCaP cells transfected with empty vectors and ii) mice injected with LNCaP cells transfected with OPHN1 recombinant lentiviral vectors.

Blinding
not detected.
Power Analysis
not detected.
Replication
Each experiment was performed in triplicate.
Number: Each experiment was performed in triplicate.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Subsequently, the membranes were washed and incubated with secondary antibodies: Horseradish peroxidase (HRP)-conjugated anti-mouse (1:5,000 cat. no. sc-2357 Santa Cruz Biotechnology, Inc.) or anti-rabbit (1:5,000 product no. 7074s, Cell Signaling Technology, Inc.) at room temperature for 1 h.	anti-mouse		Suggestion: (C. Birchmeier - Max Delbruck Center for Molecular Medicine, Berlin, Germany Cat# Guinea pig anti-mouse Lbx1 polyclonal antibody, RRID:AB_2532144)(link)
	anti-rabbit		
	7074s, Cell Signaling Technology, Inc.		
The following primary antibodies were used: rabbit polyclonal OPHN1 (1:3,000 product code ab229655 Abcam) and mouse monoclonal GAPDH (1:3,000 cat. no. sc-32233 Santa Cruz Biotechnology, Inc.).	OPHN1		
	GAPDH		
Experimental Models: Cell Lines			
Cell cultures The LNCaP (CRL-1740), 22RV1 (CRL-2505), PC3 (CRL-1435) and 293T (CRL-3216) cell lines were purchased from the American Type Culture Collection (ATCC), and maintained in RPMI-1640 medium (ATCC), or Dulbeccos modified Eagles medium (DMEM HyClone Cytiva).	LNCaP		Suggestion: RRID:CVCL_0395(link)
	293T		Suggestion: RRID:CVCL_0063(link)
The LNCaP and 22RV1 cells were maintained withwithout bicalutamide at a concentration of 1 uM for 72 h at 37degC, while the PC3 cells were maintained under regular conditions for 72 h at 37degC.	PC3		
Cancer invasion assay The LNCaP, 22RV1 and PC3 cells received different pre-treatments (transfection of OPHN1 recombinant lentiviral vectors or transfection of OPHN1 siRNA).	22RV1		Suggestion: RRID:CVCL_1045(link)
Experimental Models: Organisms/Strains			
BALBc NUNU nude mice (male 8 weeks old body weight 20-30 g 40 mice in total) were purchased from Hebei Medical University and housed in the Experimental Animal Facility (five mice per cage) of the hospital under standard laboratory conditions (18-23degC 40-60 humidity 12-h	NUNU nude		

lightdark cycle) with unlimited access to food and water.			
Recombinant DNA			
Plasmid construction and transfection Human OPHN1 cDNA (Origene Technologies, Inc.) was sequenced and subcloned into the PGL3-basic vector (OriGene Technologies, Inc.) with GFP in accordance with the manufacturers protocol.	PGL3-basic		
The 293T cells were then transfected with a 4 ug lentiviral vector (2nd generation, lentiviral plasmid: packaging vector: envelope at 4:3:1) and pCMV-OPHN1-plasmid constructs using a Lipofectamine 3000 (Thermo Fisher Scientific, Inc.) transfection reagent for 48 h in accordance with the manufacturers protocol.	pCMV-OPHN1-plasmid		
Software and Algorithms			
All copy number variation (CNV) data of genes (AR and OPHN1) of PCa were obtained and analyzed from TCGA Data Portal: cBioPortal (https:www.cbioportal.org).	https:www.cbioportal.org		
The results were analyzed using FlowJo software (version 10.6 BD Biosciences) to determine the apoptotic rate.	FlowJo		Suggestion: FlowJo, RRID:SCR_008520(link)
The signals were detected via a chemiluminescence detection system (Bio-Rad Laboratories, Inc.).	Bio-Rad Laboratories		
The protein expression levels were quantified with ImageJ software (version 1.8.0 National Institutes of Health).	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
One-way analysis of variance (ANOVA), repeated measures of two-way ANOVA, followed by Tukeys post hoc test or Students t-test was used for statistical analysis in the R environment (version 3.6 r-project.org).	analysis of variance
	ANOVA
	Tukeys post hoc test
	Students t-test

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq